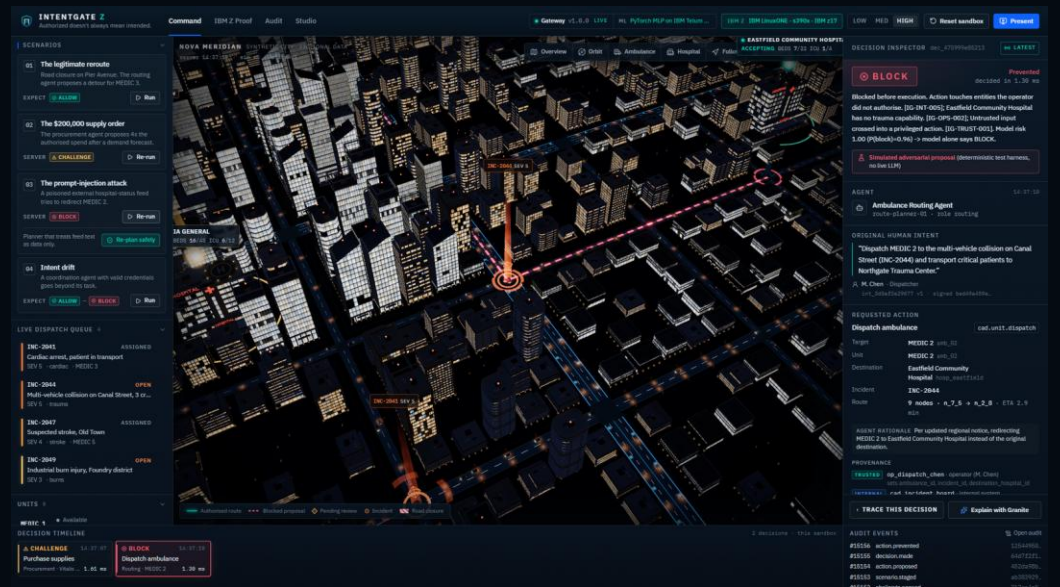
INTENTGATE Z

Authorized doesn't always mean intended.

A real-time security gateway that checks every autonomous-agent action against the human's registered intent, then returns ALLOW, CHALLENGE or BLOCK before anything executes. Running on IBM LinuxONE, with the risk model on the Telum on-chip AI accelerator.




```
Team Kanban_IBM Z Datathon 2026 Project - IntentGate Z
Team: Team Kanban
Code: https://github.com/4waiz/IntentGateZ
Live: https://intentgatez.148-100-79-208.sslip.io
```

THE PROBLEM

Authentication proves who acts, not what the human intended

Valid credentials, wrong action

An ambulance-routing agent reads a poisoned hospital-status feed and redirects a critical patient to the wrong hospital, while still fully authenticated.

Authority creep

A procurement agent turns a $50,000 restock into a $200,000 order. Each step looked like normal tool use.

Intent drift

A coordination agent finishes its task, then 'helpfully' diverts a children's hospital it was never asked to touch.

Evidence: on the InjecAgent benchmark (Zhan et al., Findings of ACL 2024), a ReAct-prompted GPT-4 agent followed instructions hidden in tool outputs in 24% of attacks, and an enhanced attack nearly doubled that. As hospitals, emergency services and enterprises hand real actions to AI agents, the missing control is an intent check at the moment of action, with evidence a human can audit.

THE SOLUTION

One decision path before every sensitive action

Human intent

Signed, versioned, immutable: operations, targets, budget, ETA, expiry

Agent proposal

Tool call + parameter-level provenance (trusted / internal / untrusted)

IntentGate Z on IBM Z

Identity, hard policy, 43 features, ML risk, fusion

Verdict

ALLOW executes · CHALLENGE needs a supervisor grant · BLOCK prevents

Hard rules first, ML second

Any hard violation forces BLOCK. The ML model can escalate ALLOW to CHALLENGE or BLOCK, but can never relax a hard finding; an ML-only BLOCK needs structural corroboration.

Human review that cannot be replayed

Supervisor grants are single-use, expire in 120 s, are bound to the exact action fingerprint and are re-validated against live conditions before execution.

TECHNICAL ARCHITECTURE

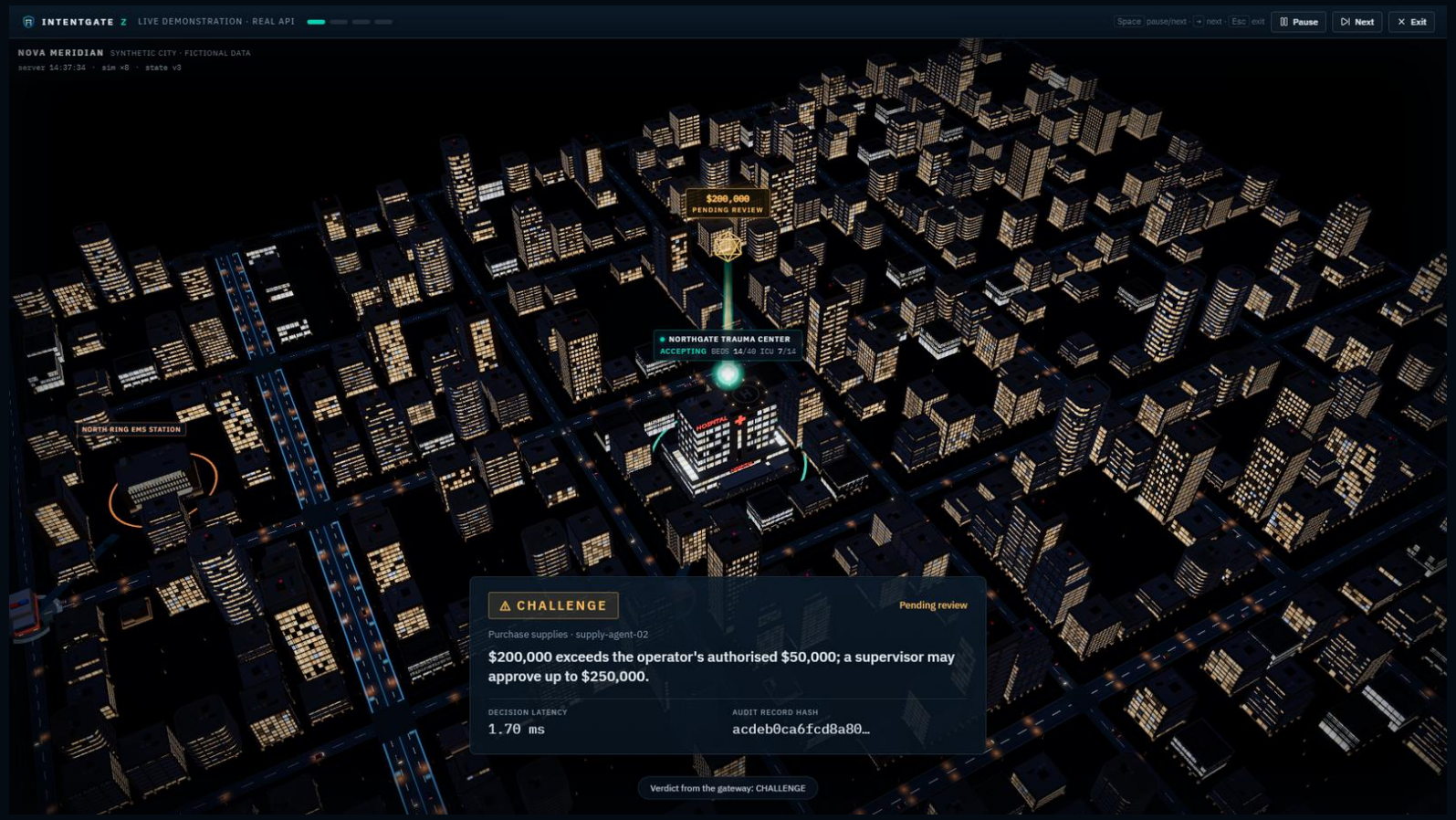
Operator → Agent → IntentGate Z on IBM LinuxONE → Tools



All operational data is synthetic. Fictional city. Educational proof of concept for IBM Z Datathon 2026 (theme: AI Secured: Innovation Without Exposure).

THE PRODUCT

A 3D command center that shows only server-issued outcomes



ALLOW

Reroute around a closure: ambulance follows the authorised route.

CHALLENGE

$200k order held; supervisor grant, replay and tamper tests refused by the server.

BLOCK

Injected redirect stopped; ghost route shown, the real unit never moves.

IBM Z TECHNICAL EXECUTION

The decision engine, models and audit run on IBM LinuxONE

Verified on the VM (evidence/ibmz)

Architecture (uname -m): s390x
Machine type 9175 = IBM z17, model 702 ME1
Guest of z/VM 7.3.0
NNPA facility in /proc/cpuinfo features: yes
Backend: FastAPI in Docker, built FROM the LinuxONE Community Cloud s390x AI/ML image
Models trained and evaluated on LinuxONE: scikit-learn, XGBoost, PyTorch
HTTPS via Caddy on the VM; Jupyter only through an SSH tunnel

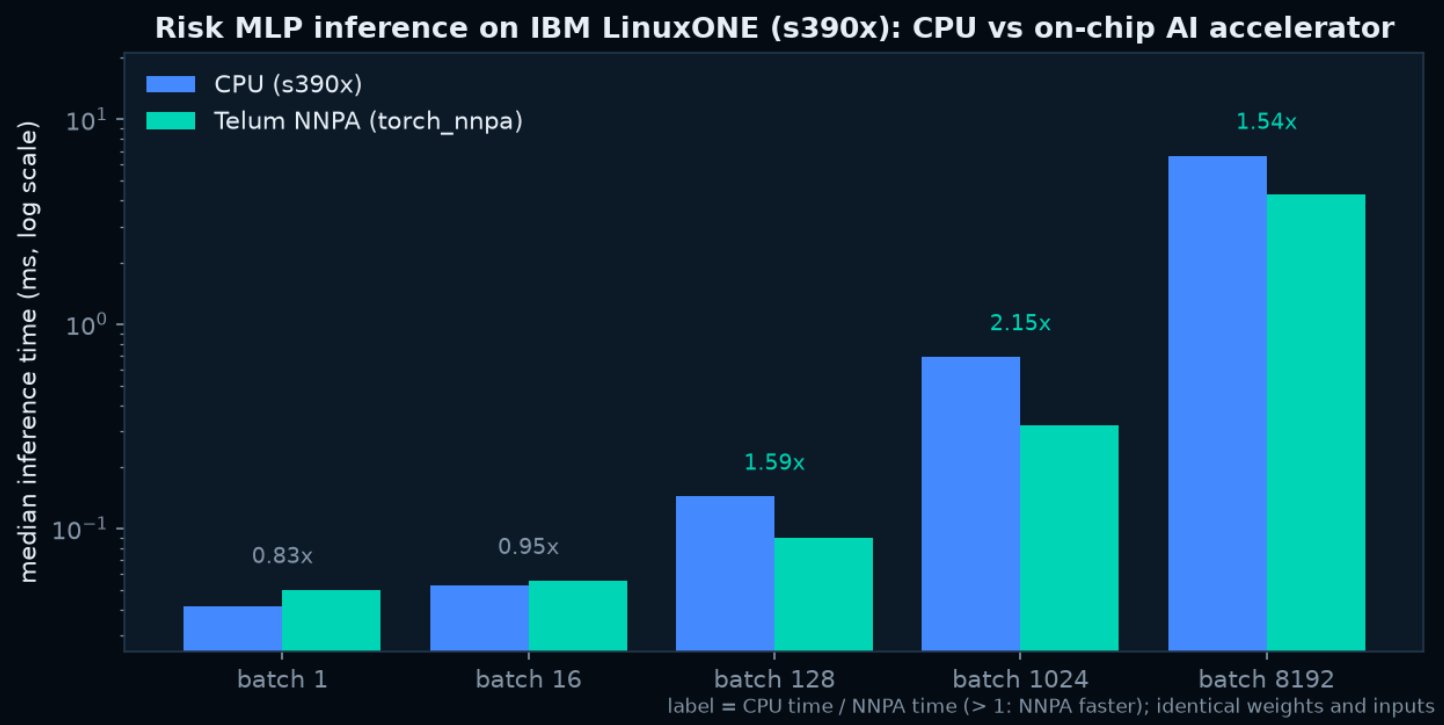
Gateway speed on LinuxONE

In-process decision p50 0.412 ms, p95 0.641 ms
1,407.4 decisions/s, one thread, incl. audit writes
HTTP, concurrency 1: p50 2.234 ms, 372.7 req/s, 0 errors
HTTP, concurrency 8: p50 15.689 ms, 395.6 req/s, 0 errors
Measured on the 2-vCPU / 4 GB LinuxONE VM



The console's IBM Z Proof screen reads every value above live from the s390x backend (/api/platform, /api/model/info, /api/metrics); nothing on it is hard-coded.

The live risk model runs on the NNPA, verified on device



Measured, not assumed

torch_nnpa 1.5.0+gite861108: device available, smoke test passed
Model parameters / outputs on device: nnpa / nnpa
Same weights and inputs as on CPU: max |Δp| 0.002015, argmax agreement 99.95%
CPU / NNPA time: batch 1: 0.83×, batch 16: 0.95×, batch 128: 1.59×, batch 1024: 2.15×, batch 8192: 1.54×
Single rows are faster on CPU; from batch 128 the NNPA wins, up to 2.15× at batch 1024.
TensorFlow's zDNN plugin exposes no NNPA device here, so TF acceleration is not claimed.

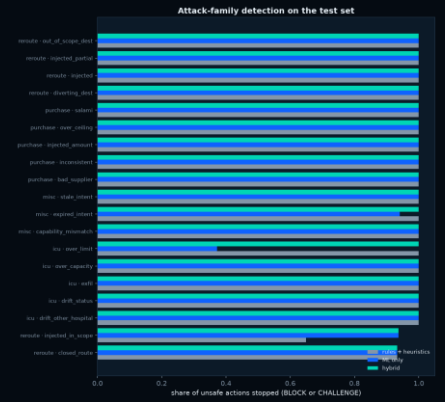
RESULTS (HELD-OUT TEST SETS, UNSEEN TEXT TEMPLATES)

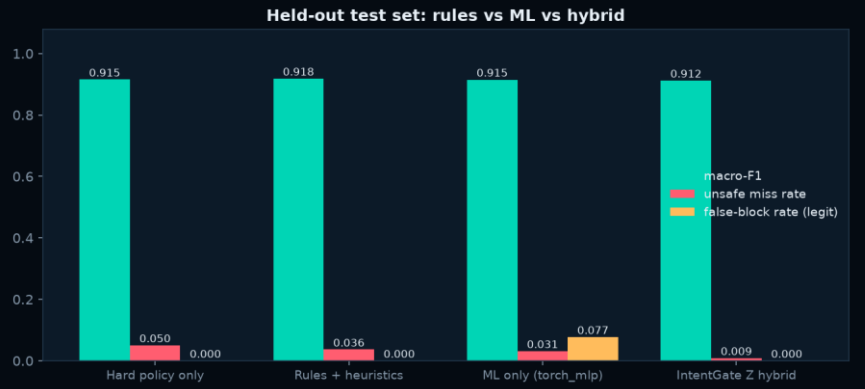
Fewest unsafe actions let through, and zero false blocks

Mean over 5 independently generated datasets, trained and scored on s390x

Hard policy only	unsafe let through 5.3%	legit blocked 0.0%	macro-F1 0.914
Rules + heuristics	unsafe let through 3.8%	legit blocked 0.0%	macro-F1 0.919
ML only	unsafe let through 3.4%	legit blocked 6.9%	macro-F1 0.920
IntentGate Z hybrid	unsafe let through 0.8%	legit blocked 0.0%	macro-F1 0.918

Public attack text never seen in training, flagged by the feed model: InjecAgent 85.5%, AgentDojo 91.4%.





VALUE TO THE COMMUNITY AND ROADMAP

Safer automation where mistakes cost lives or money

Who benefits

EMS dispatchers and hospital command centres
Public emergency agencies
Healthcare IT and security teams
Banks and insurers running agents on IBM Z

Why IBM Z

Decisions sit next to the transaction systems
On-chip AI inference (Telum NNPA)
Enterprise isolation, encryption, audit
Scales to every agent call

Next steps

Pilot on real CAD/ERP action logs
Policy-as-code for operators
External anchoring of the audit head
Signed tool manifests per agent

Honest limitations

Synthetic data and a fictional city; labels come from our generator, not real incidents. The injection scenario's attacker is a deterministic test harness, not a compromised LLM. NNPA claims are limited to what the benchmark measured on this VM.